

ABHINAV BIRRU

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Education

Indian Institute of Technology, Guwahati

Nov 2020 – May 2024

Bachelor of Technology in Electronics and Communication Engineering

CGPA: 8.67

Experience

AQR Capital India Services LLP

July 2024 - Present

Software Developer - QRD

Bengaluru, Karnataka

Historical Factor Refresh & Signal Library Consolidation

- Consolidated the research and production factor trees into a single unified tree, then drove the annual historical refresh for the resulting 250K+ factor library - recomputing every leaf-level RAW and derived signal back to 1980s.
- Designed a backfill and stitching methodology for datasets like Betas, Universes whose production history is shorter than research, identified analogous research datasets, validated statistical equivalence, and stitched histories.
- Reconciled divergent factor-computation logic between research and production codebases into a single canonical implementation, extended the production tree to cover research-only signals - coordinating with signal owners to migrate, validate, sign-off each factor, removing a long stanfing source of signal drift.
- Diagnosed ~ 40 distinct failure modes across 600+ RAW factors (data-gaps, cross-environment logic differences) and built an owner-routed triage workflow to resolve the failures.

Research-Production Environment Consolidation

- Owned end-to-end consolidation of research and production across 3 workflows - data loading, universe updates, and factor computation/return loading - eliminating redundant research pipelines and redirecting analysts onto stable, signed-off production data.
- Stitched research history onto the corresponding production items, establishing production as the single source of truth.
- Cut monthly research factor computation down to once-a-year annual refresh by routing day-to-day needs through production outputs, removing a recurring compute and support burden.
- Extended production factor-return loading to cover non-leaf research factors; migrated official research factor returns from SQL Server to Postgres and generalized the prod return-fetch API with an EnvironmentMode parameter to support both production and research calls using same code.

Infrastructure Modernization & Production Engineering

- Contributed to migration of a legacy Python 2 Windows analyisics stack to a Python 3 Linux runtime on AWS Batch; refactoring the monolith codebase into modular cloud-native libraries.
- Built and maintained Docker containers for reproducible execution across dev, research, and production environments.
- Migrated batch file storage from network shared drive pickle files to Parquet on AWS with S3 versioning, achieving ~ 70 – 80% reduction in storage footprint and eliminating a class of bugs caused by pickle protocol incompatibility between Python 3.6 and 3.11 environments; Parquet’s columnar format, embedded schema, and language-agnostic readability also made the data usable beyond python.

AQR Capital India Services LLP

May 2023 - July 2023

Software Engineering Intern

Bengaluru, Karnataka

Analytics & Monitoring Dashboard (Python)

- Built a Python/Flask metrics library to ingest production process metrics into Elasticsearch, Kibana based dashboards for real-time monitoring of the metrics.
- Automated SQL data collection using StoneBranch Jobs via scheduled triggers.

Projects

IITG-Campus-OLX 

Aug.2023 - Nov 2023

Group Project, Coding Club, IITG

- E-Commerce Web App:** Designed and built a React JS web application for the IITG community supporting buying, selling and renting of products within campus, with a responsive UI for listing and discovery.
- Authentication and Media Storage:** Integrated Firebase for user authentication and used used Firebase Storage to securely upload, serve and manage product images.
- Search Backend:** Integrated Elasticsearch to index peroduct metadata and serve low-latency product search across the catalog, improving discoverability over a basic SQL LIKE-based search.
- User Profiles & Listing Management:** Implemented per-user profiles with listing creation, editing, status tracking.

Library Management System 

Jun.2022 - July 2022

Personal Project

- Flask Web Application:** Designed and implemented a **Flask** and **MySQL**-based web application for efficient library book management, catering to both users and librarians.
- Role-Based Interfaces:** Developed separate, user-friendly interfaces for librarians (for book inventory management) and users (for book transactions and personalization).
- User Authentication & Features:** Implemented secure user authentication and functionalities, enabling users to add, issue, reserve books, maintain personal bookshelves, and view transaction histories.

Technical Skills

Languages: C, C++, SQL, Python
Tools/Frameworks: AWS, JupyterHub, Jenkins, Docker, Flask, NumPy, Pandas, Parquet, Linux
Databases: MSSQL, Elasticsearch, Firebase, Postgres

Achievements

- Flipkart Grid 5.0:** Among the Top 0.5% teams shortlisted in the Online quiz out of 3,97,375 teams
- JEE Advanced 2020** Secured an All India Rank 4656 among 1,15,000 candidates.
- JEE Mains 2020** Secured an All India Rank 651 among 11,20,000 candidates.

Leadership / Extracurricular

Department Placement Coordinator

May 2023 – July 2024

Centre For Career Development

IIT Guwahati

- Responsible for managing placement process for EEE department.